



**Department of Energy Science & Engineering
Indian Institute of Technology Bombay**

**EN639 Renewable Energy Integration
Project Report**

Development of Irrigation Feeder with Solar PV Powered Pumps

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Chapter 1

Introduction

India's agriculture sector plays a vital role in the country's economy, contributing significantly to GDP and providing livelihoods to a large portion of the population. It supports food security for over a billion people and supplies raw materials to various industries. Despite modernization in certain areas, a major part of Indian agriculture still depends on monsoon rainfall, making it vulnerable to climate variability. To enhance productivity and ensure sustainability, reliable access to water and energy has become increasingly important.

Irrigation feeders are dedicated electrical distribution lines designed specifically to supply power to agricultural loads, primarily water pumps used for irrigation. These feeders help in segregating agricultural demand from other types of loads, allowing better load management and power quality control. Pumping loads connected to these feeders are typically inductive in nature and vary depending on factors such as crop type, groundwater level, and seasonal requirements. However, these loads often lead to issues such as peak demand stress, voltage fluctuations, and high transmission losses due to their dispersed and unregulated usage.

In recent years, the integration of solar energy for agricultural pumping has emerged as a sustainable and efficient solution. Solar-powered irrigation systems reduce dependence on grid electricity and diesel, thereby lowering operational costs and environmental impact. By utilizing abundant solar resources, especially in rural and remote areas, these systems provide a reliable and decentralized energy source for farmers. Additionally, coupling solar generation with irrigation feeders can improve grid stability, reduce peak load pressures, and promote cleaner energy usage, aligning with India's renewable energy goals.

This report discusses a small network of solar PV powered pumps connected to irrigation feeder. The data used in this study has been derived from the Maharashtra state load profile, which provides a detailed representation of regional electricity demand characteristics. In addition, the irrigation load component has also been obtained for the same state, ensuring consistency in demand modeling and maintaining realistic representation of agricultural consumption patterns within the system.

Chapter 2

Literature Review

Reference Framework for Feeder

The literature review for designing irrigation feeder is based on the methodology presented in [?], where the authors developed a reduced electrically equivalent model of the **IEEE European Low Voltage Test Feeder**. Their study focuses on simplifying a detailed distribution network into a computationally efficient model while preserving its essential electrical characteristics, such as voltage profile, power flow behavior, and load representation. This reduced modeling approach is particularly useful for time-series simulations and power system analysis where full-scale network modeling becomes complex and computationally intensive.

In the present study, a similar approach has been adopted for the development of the irrigation feeder model. The reference paper provides the foundation for feeder reduction, bus representation, and load aggregation techniques used in this work. Based on this framework, a simplified grid-feeder-load system has been constructed to represent the irrigation distribution network while maintaining its operational behavior for simulation purposes.

Solar Irradiance data

The solar irradiance data used in this study has been obtained from the National Solar Radiation Database (NSRDB) 2 international dataset available through the National Renewable Energy Laboratory (NREL) platform. The NSRDB is a widely recognized and publicly accessible database that provides high-resolution, satellite-derived solar radiation and meteorological data for multiple geographical regions, including India. It offers time-series data of key solar irradiance components such as Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI), typically at half-hourly or hourly intervals, enabling detailed temporal and spatial analysis of solar energy potential.

Irrigation feeder Load profile

The irrigation feeder load profile used in this study is developed using national-level electricity consumption data available from the India Climate and Energy Dashboard (ICED) of NITI Aayog 3. The dashboard provides hourly demand curves and category-wise electricity consumption patterns for the power distribution sector, which are useful for understanding daily and seasonal variations in electricity demand across agricultural and rural feeders. It includes information on national electricity demand, hourly load curves, and consumption trends that support feeder-level demand estimation.

Based on this reference, the overall irrigation feeder demand curve in this work is derived by adapting the available national load curve data to represent agricultural pumping demand. The detailed procedure for converting the available data into the irrigation feeder load profile, including load allocation and scaling methodology, is explained in the methodology section. This approach ensures that the developed feeder model reflects realistic operating conditions while maintaining consistency with observed electricity consumption patterns.

Chapter 3

Equivalent Modeling of Feeder, Load System and Grid

3.1 Reduced Irrigation Feeder Model

The modeling approach and network configuration adopted in this work are inspired by the study conducted [?]. Their work presents a reduced electrically equivalent model of the **IEEE European Low Voltage Test Feeder**, focusing on simplifying detailed distribution networks while preserving key electrical characteristics. This enables efficient time-series power flow analysis and computational modeling.

The feeder network has been further reduced to make the computational analysis feasible in MATLAB. The resulting simplified model consists of 6 no-load buses and 6 Pumping loads, while still preserving the essential electrical characteristics required for the study.

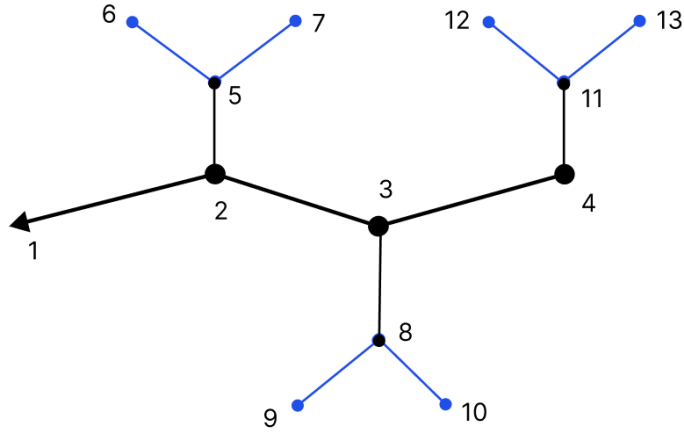


Figure 3.1: Feeder Network for Simulation (Reduced)

Bus Number(s)	Bus Type
1	Substation (Grid)
2, 3, 4, 5, 8, 11	No-load Bus
6, 10, 12	Pump Load (5 HP)
7, 9, 13	Pump Load (7.5 HP)

Table 3.1: Classification of Buses in the Irrigation Feeder Network

By performing the calculations, a reduced irrigation feeder model is obtained for the simulation. The updated line resistance and reactance values represent the actual electrical characteristics of the feeder more accurately, enabling effective and realistic load flow analysis and performance evaluation of the solar PV powered irrigation pump network.

Lines	Node A	Node B	Phases	Length (m)	Line Code
1	1	2	ABC	21.867341	4c-70
2	2	5	ABC	5.9049	2c-16
3	2	3	ABC	12.6224	4c-70
4	5	6	ABC	5.3513	2c-16
5	5	7	ABC	6.747143	2c-16
6	3	8	ABC	5.7265	2c-16
7	3	4	ABC	21.15538	4c-70
8	8	9	ABC	6.1526	2c-16
9	8	10	ABC	6.44409	2c-16
10	4	11	ABC	8.2583	2c-16
11	11	12	ABC	4.0366	2c-16
12	11	13	ABC	4.0366	2c-16

Table 3.2: Line Parameters

Line Code	R	XL	Unit
4c-70	0.446	0.071	per km
2c-17	1.15	0.088	per km

Table 3.3: Line Codes

Calculation of Line Parameters

The resistance and reactance of each line are calculated using:

$$R = R_{per\ km} \times \frac{L}{1000}, \quad X_L = X_{L,per\ km} \times \frac{L}{1000}$$

Sample Calculation for Line 1:

$$L = \frac{21.867341}{1000} = 0.021867km$$

$$R = 0.446 \times 0.021867 = 0.00975\ \Omega$$

$$X_L = 0.071 \times 0.021867 = 0.00155\ \Omega$$

The final calculated values of resistances and reactances for the different transmission lines in the reduced feeder network are presented below. These parameters have been derived to accurately represent the electrical characteristics of each line segment in the simplified model.

Line	Length (m)	Code	R (Ω)	X_L (Ω)
1	21.867	4c-70	0.00975	0.00155
2	5.9049	2c-16	0.00679	0.00052
3	12.6224	4c-70	0.00563	0.00090
4	5.3513	2c-16	0.00615	0.00047
5	6.7471	2c-16	0.00776	0.00059
6	5.7265	2c-16	0.00659	0.00050
7	21.1554	4c-70	0.00943	0.00150
8	6.1526	2c-16	0.00708	0.00054
9	6.4441	2c-16	0.00741	0.00057
10	8.2583	2c-16	0.00950	0.00073
11	4.0366	2c-16	0.00464	0.00036
12	4.0366	2c-16	0.00464	0.00036

Table 3.4: Calculated Line Resistance and Reactance

3.2 Grid

The Grid model is built in a three-phase configuration to capture realistic power system behavior under unbalanced and time-varying agricultural loads.

The grid is connected to the downstream network through a controlled three-phase interface, ensuring proper representation of supply conditions for the irrigation feeder model. This configuration allows the simulation of realistic grid behavior under varying load demand conditions in the feeder.

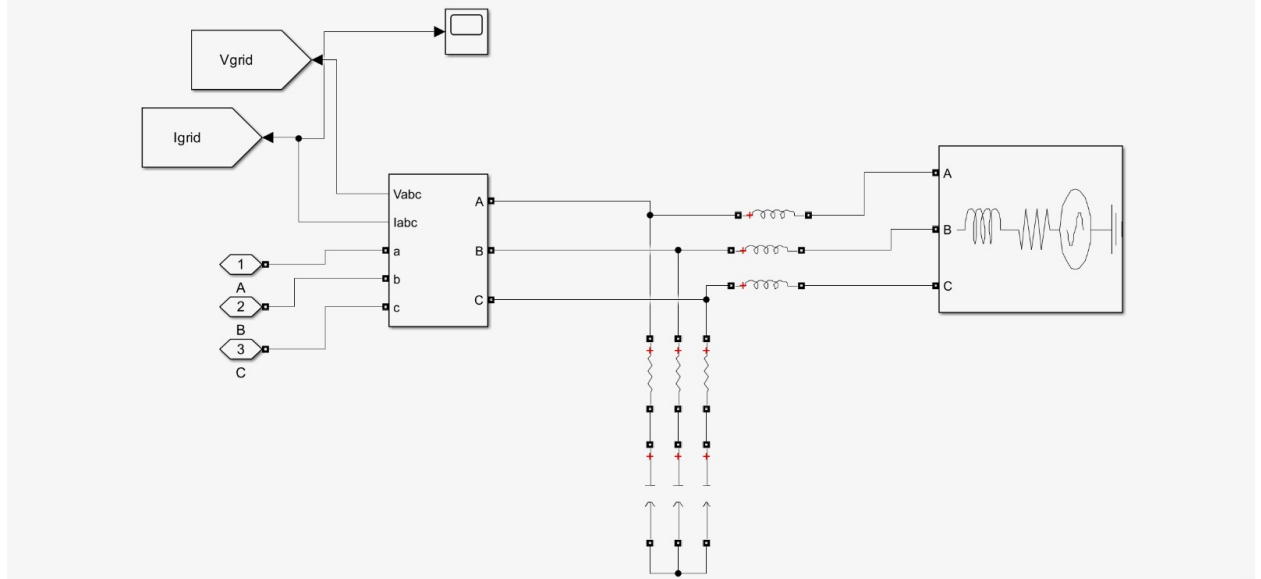


Figure 3.2: Grid (source network)

The model shown represents the grid supply-side (source network) of the developed grid-feeder-load simulation system. It is implemented to provide a stiff three-phase voltage source (V_{grid}) that supplies power to the downstream irrigation feeder network (shown in section 3.1).

The L and C for the grid and the inverter output were designed, taking into consideration the maximum power consumption at the load.

3.3 Pumping Load

The pumping load attached to feeder are selected based on commonly used irrigation practices and commercially available equipment suitable for agricultural applications.

Pump Type	No. of Pumps	Equivalent Power (kW)
5.4 HP Pumps	3	4 kW each
10 HP Pumps	3	7.5 kW each

Table 3.5: Pump Rating

AC Load Calculation

The total AC load of the irrigation pumping system is calculated as:

$$\begin{aligned}P_{AC} &= (4 + 7.5) \times 3 \\&= 11.5 \times 3 \\&= 34.5 \text{ kW}\end{aligned}$$

DC Capacity Calculation

Considering the inverter loading ratio, the DC capacity is taken as 1.25 times () the AC load:

$$\begin{aligned}P_{DC} &= P_{AC} \times 1.25 \\&= 34.5 \times 1.25 \\&= 43.125 \text{ kW}\end{aligned}$$

Parameter	Value
Total AC Load	34.5 kW
DC Capacity (PV System Size)	43.125 kW

Table 3.6: Final System Capacity Summary

Hence, the design of photovoltaic system capacity should **43.125 kW (DC)** minimum.

Chapter 4

Solar PV System Configuration

PV System Design

The photovoltaic (PV) system has been designed based on the required DC capacity of the irrigation pumping system. The total required system capacity is **43.125 kW**. However, considering system losses such as wiring losses, inverter losses, and environmental derating factors, the system is designed for a slightly higher capacity of **45 kW** to ensure reliable and efficient operation under varying conditions.

The selected PV module is a **600 Wp** Waaree ELITE SERIES N-TYPE module.

PV Array Sizing

The total number of PV modules required is calculated as:

$$N_{modules} = \frac{P_{DC}}{P_{module}} = \frac{45 \times 10^3}{600} = 75 \text{ modules} \quad (4.1)$$

Parameter	Value
Total DC Capacity	45 kW
PV Module Rating	600 Wp
Total Number of Modules	75

Table 4.1: PV System Sizing Parameters

Array Configuration

The PV modules are arranged in a string configuration to ensure proper voltage and current matching with the inverter.

Figure 4.1: PV Array Configuration

Configuration Parameter	Value
Number of Strings	5
Modules per String	15
Total Modules	75
Module Rating	600 Wp

Table 4.2: PV Array Configuration

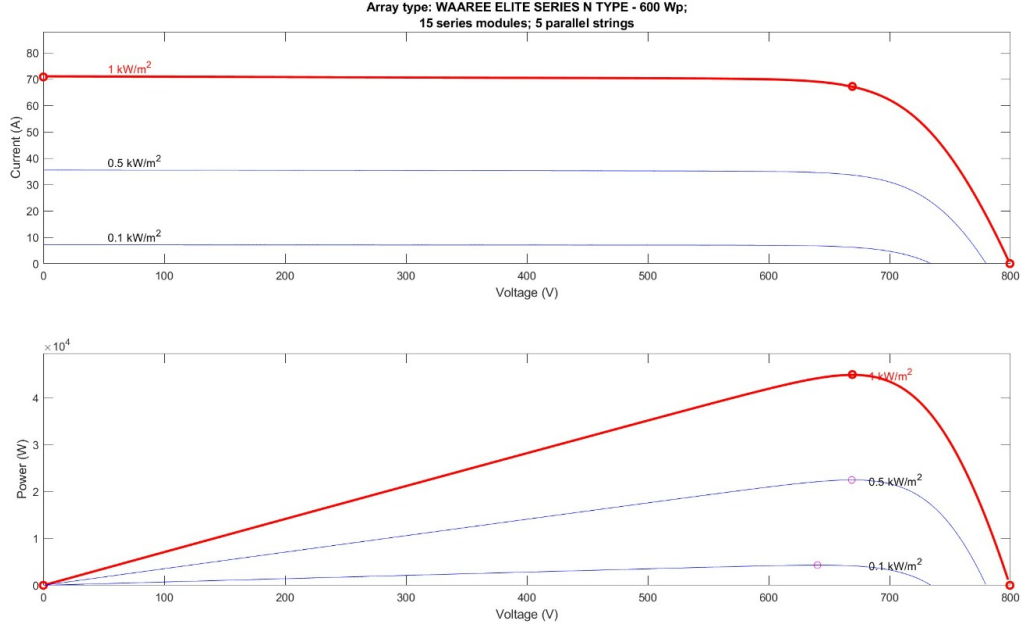


Figure 4.2: PV Module Characteristics

Figure 4.2 shows the I–V (Current–Voltage) and P–V (Power–Voltage) characteristics of the designed PV array under different solar irradiance levels. It can be observed that as solar irradiance increases, the output current increases significantly, while the open-circuit voltage changes only slightly. At standard test conditions of 1 kW/m^2 , the array delivers a maximum current of nearly **70 A** and an open-circuit voltage close to **800 V**.

The P–V characteristics indicate that the maximum power point (MPP) occurs around **650–700 V**, where the array (15x5) produces approximately **45 kW**, matching the designed system capacity. At lower irradiance levels, both the output current and maximum power reduce proportionally, which reflects the direct dependence of PV output on solar radiation availability.

These characteristics confirm that the selected module configuration is suitable for the irrigation pumping application and provides stable operation within the inverter operating voltage range while ensuring efficient energy extraction under varying weather conditions.

Since the PV array output voltage is around 700 V, while the inverter requires a higher and stable DC input voltage for efficient operation, a DC–DC boost converter is added after the PV array to step up and regulate the voltage before feeding it to the inverter.

Chapter 5

Overall System Configuration

System Description

The overall system is developed in MATLAB/Simulink to model a grid-connected solar photovoltaic (PV) based irrigation feeder network incorporating seasonal load variations. The simulation environment integrates three principal subsystems: a solar PV generation chain with power electronic converters (boost converter and a inverter), a grid source model, and a reduced agricultural distribution feeder with time-varying load profiles. Together, these subsystems allow time-series power flow analysis under realistic seasonal operating conditions representative of the Maharashtra agricultural context.

5.1 PV Generation and MPPT

The PV generation subsystem is modelled using hourly irradiance (GHI) and ambient temperature data obtained from the NSRDB dataset for three seasonal conditions Summer, Winter, and Monsoon corresponding to the climatic characteristics of Maharashtra (Section 6.1). These environmental inputs are routed through a switching mechanism that activates the appropriate seasonal data block during simulation, enabling representative time-series analysis without the computational cost of a full annual simulation. The PV array (45 kW, 75×600 Wp modules in a 15-series $\times$ 5-parallel configuration; Section 4) feeds its output to a Maximum Power Point Tracking (MPPT) controller. The MPPT algorithm (figure 5.1) continuously adjusts the array operating point to extract the maximum available power under the prevailing irradiance and temperature, which is particularly important during the Monsoon season when rapid cloud-induced irradiance transients cause frequent shifts in the P–V characteristic.

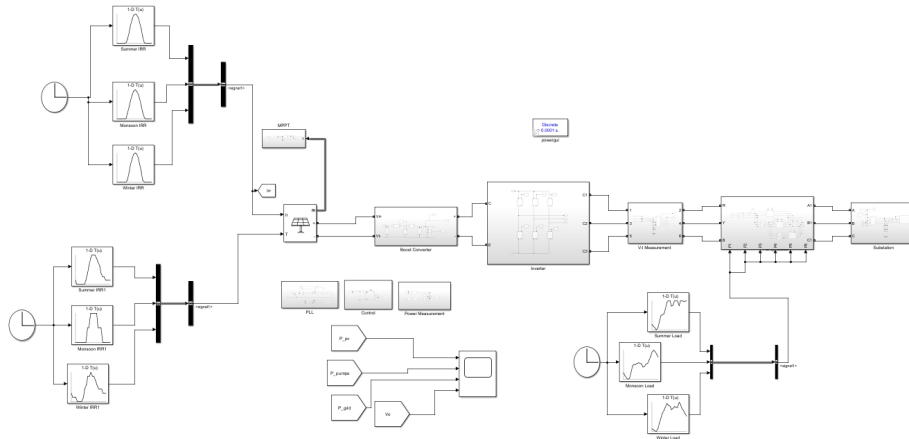


Figure 5.1: System Configuration

EN639_project_v9
 MPPT
 MATLAB Function

```

1      function Vref = fcn(P, V)
2
3      % P&O MPPT Algorithm (Improved Version)
4
5      % ---- USER PARAMETERS ----
6      Vref_initial = 750;    % Initial guess near expected MPP (~800 V)
7      Vrefmax = 900;        % Upper voltage limit
8      Vrefmin = 500;        % Lower voltage limit
9      deltaV = 1;          % Step size (can tune: 5-10 for high voltage systems)
10     epsilon = 1e-3;       % Deadband for power change
11
12     % ---- PERSISTENT VARIABLES ----
13     persistent Vold Pold Vrefold
14
15     if isempty(Vold)
16         Vold = V;
17         Pold = P;
18         Vrefold = Vref_initial;
19     end
20
21     % ---- COMPUTE DIFFERENCES ----
22     dV = V - Vold;
23     dP = P - Pold;
24
25     % ---- P&O LOGIC ----
26     if abs(dP) < epsilon
27         % Very small change → stay at same point
28         Vref = Vrefold;
29
30     else
31         if dP > 0
32             if dV >= 0
33                 Vref = Vrefold + deltaV;
34             else
35                 Vref = Vrefold - deltaV;
36             end
37         else
38             if dV < 0
39                 Vref = Vrefold + deltaV;
40             else
41                 Vref = Vrefold - deltaV;
42             end
43         end
44     end
45
46     % ---- SATURATION (IMPORTANT) ----
47     Vref = min(max(Vref, Vrefmin), Vrefmax);
48
49     % ---- UPDATE MEMORY ----
50     Vrefold = Vref;
51     Vold = V;
52     Pold = P;
53
54     end
  
```

Figure 5.2: MPPT code

5.2 DC–DC Boost Converter

The MPPT output is passed to a DC–DC Boost Converter, which steps up the PV array terminal voltage—typically 650–700 V at the maximum power point—to a higher, regulated DC bus voltage suitable for efficient VSI operation (nominal DC link voltage 800 V). The converter duty cycle is modulated by the MPPT controller output, ensuring that both maximum power extraction and DC bus regulation are achieved simultaneously. The boost converter also acts as the primary buffer between the variable PV source and the downstream inverter, limiting the rate at which DC link voltage fluctuations propagate to the AC side.

Design of inductor uses the formula:

$$L = \frac{V_{in,min} \cdot D}{\Delta I_L \cdot f_{sw}} \quad (5.1)$$

where $\Delta I_L = 30\% \cdot I_{in}$ and

$$I_{in} = \frac{P_{DC}}{V_{in,min}} \quad (5.2)$$

Design of capacitor uses the formula:

$$C = \frac{P_{DC} \cdot D}{V_o \cdot \Delta V_o \cdot f_{sw}} \quad (5.3)$$

where $\Delta V_o = 5\% \cdot V_o$

The duty ratio is calculated using:

$$D = 1 - \frac{V_{in,min}}{V_o} = 1 - \frac{500}{800} = 0.375 \quad (5.4)$$

with $f_{sw} = 5000 \text{ Hz}$

Thus, the obtained values are $L = 0.694 \text{ mH}$ & $C = 52.734 \text{ }\mu\text{F}$

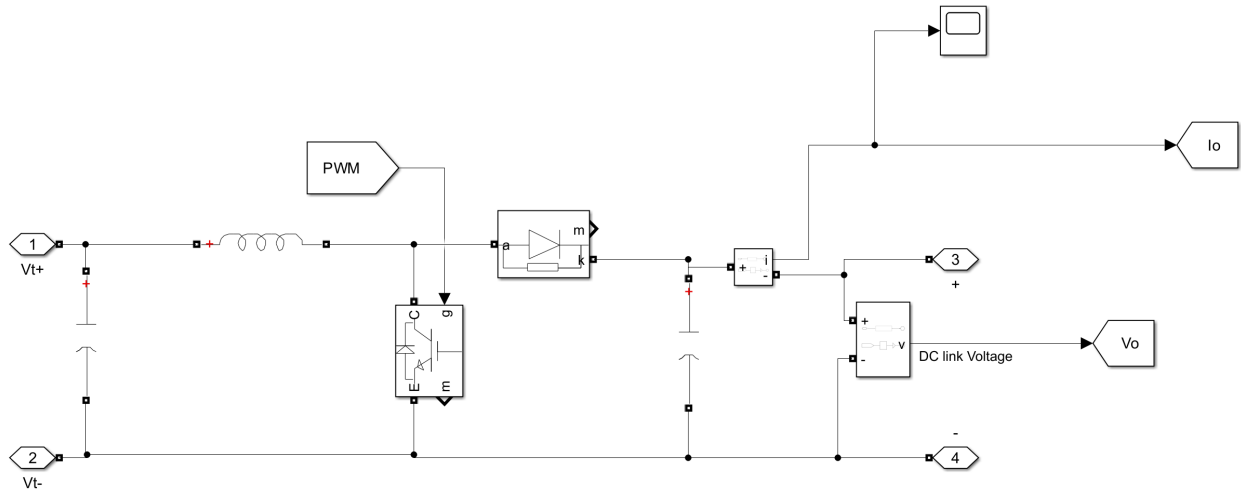


Figure 5.3: Boost converter

5.3 Inverter and Control Strategy

The regulated DC power from the photovoltaic source is converted into three-phase AC power using a voltage source inverter (VSI), which enables power transfer from the DC-link to the utility grid. For safe and efficient grid integration, the inverter output must match the grid voltage in terms of frequency, phase, and magnitude. This synchronization is achieved using a Phase Locked Loop (PLL), which continuously tracks the grid voltage angle and frequency, ensuring that the inverter remains aligned with the grid under varying operating conditions.

For control implementation, the measured three-phase grid voltages and currents in the abc reference frame are transformed into the synchronously rotating dq reference frame using the Clarke & Park transformation. In this rotating frame, the sinusoidal AC quantities appear as DC-like signals under steady-state conditions, simplifying control. This transformation also decouples the active and reactive power components, allowing independent regulation of power flow to the grid.

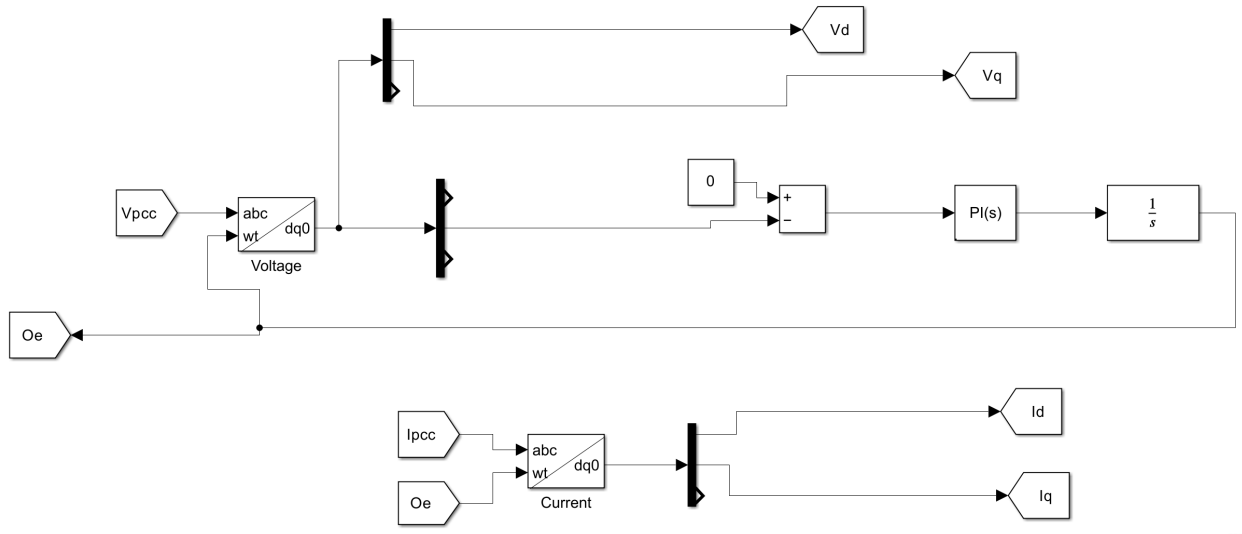


Figure 5.4: PLL Control Loop

A cascaded control structure consisting of an outer voltage loop and an inner current loop is employed for inverter control. The outer voltage loop regulates the DC-link voltage by generating the reference current required to maintain a constant DC-link voltage despite variations in solar generation or load demand. The inner current loop then ensures accurate tracking of these reference currents in the dq frame for controlled power injection. In addition, a feed-forward compensation term based on the measured grid voltage is incorporated to improve dynamic response, reduce the effect of grid disturbances, and enhance overall system stability.

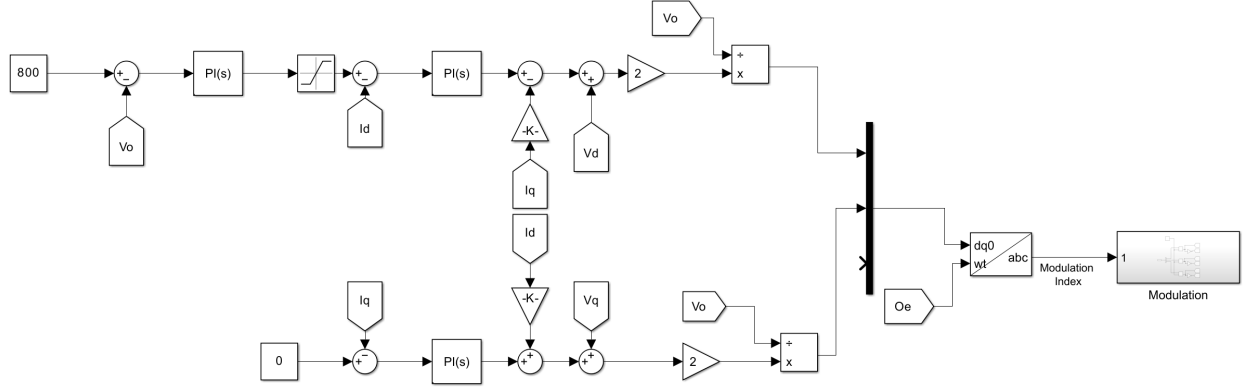


Figure 5.5: Cascaded Control Loop

5.4 Grid Source Model

The utility grid is represented as a stiff three-phase voltage source connected to Bus 1 (the substation bus) of the reduced irrigation feeder. This configuration reflects the normal operating condition of a rural feeder supplied from a distribution substation, where the grid voltage magnitude and frequency are assumed to remain nominally constant despite load and generation fluctuations within the feeder. The grid supplies or absorbs the real power difference between the total pumping load and the instantaneous PV generation, acting as the balancing entity in the power flow.

To minimize current and voltage ripples injected into the grid, an LC filter is employed at both the inverter output and the grid interface. The inductor primarily limits the high-frequency current ripple, while the capacitor smoothens the voltage by filtering switching harmonics.

A three-phase LC filter is used at the grid source side to smooth the voltage and current waveform before connection to the feeder. The filter parameters are selected for each phase.

Design of inductor comes from the formula:

$$L = \frac{V_{DC}}{6 \cdot f_{sw} \cdot \Delta I} \quad (5.6)$$

where $\Delta I = 10\% \cdot I_{rated}$ and

$$I_{rated} = \frac{P_{AC}}{\sqrt{3} \cdot V_{LL}} \quad (5.7)$$

Design of capacitor comes from the formula:

$$C = 0.05 \cdot C_b = \frac{0.05}{\omega_g Z_b} \quad (5.8)$$

where ω_g = grid angular frequency and

$$Z_b = \frac{V_{LL}^2}{P_{AC}} \quad (5.9)$$

Thus, the obtained values are $L = 1.339 \text{ mH}$ & $C = 34.324 \text{ }\mu\text{F}$

To verify these values,

$$f_r = \frac{1}{2\pi\sqrt{LC}} = 741.76 \text{ Hz} \quad (5.10)$$

Thus, the condition $10 \cdot f_g < f_r < 0.1 \cdot f_{sw}$ is satisfied.

To design the damping resistance in series with the capacitor:

$$R_d = \frac{1}{3 \cdot \omega_r \cdot C} = 2.084 \text{ }\Omega \quad (5.11)$$

5.5 Distribution Feeder and Seasonal Load Model

The AC output of the VSI is injected into the reduced irrigation feeder network (Section 3.1), which consists of 13 buses—one substation bus, six no-load buses, and six pump load buses—interconnected by line segments whose resistance and reactance parameters have been calculated from standard line codes (Section 3.1, Table 3.4). The feeder topology preserves the essential electrical characteristics of a real agricultural distribution feeder while remaining computationally tractable for time-series simulation in MATLAB/Simulink. The six pump load buses are assigned seasonal active power demand profiles derived from the Maharashtra state hourly load curve scaled by season-specific agricultural load fractions (Section 6.2): Summer (0.30), Winter (0.20), and Monsoon (0.15). The simulations are carried out for 24 hours, separately for the three seasons to analyze the performance of the solar PV powered irrigation feeder under different seasonal irradiance and agricultural load conditions.

Chapter 6

Irradiance Data and Feeder Load Profile

6.1 Solar Irradiance Data

The solar irradiance data used in this study has been obtained from the National Solar Radiation Database (NSRDB) international dataset, provided by the National Renewable Energy Laboratory (NREL). The dataset contains high-resolution, satellite-derived solar radiation and meteorological data for the selected location in Maharashtra, India, with half-hourly time resolution.

The selected site corresponds to the geographical coordinates of approximately latitude 19.75°N and longitude 75.71°E , representing a typical agricultural region in Maharashtra suitable for irrigation feeder analysis. In addition to solar irradiance, ambient temperature data is also extracted from the dataset since temperature significantly affects photovoltaic (PV) module performance and efficiency.

The key irradiance parameter used in this study is the Global Horizontal Irradiance (GHI), which serves as the primary input to the PV array model in MATLAB/Simulink. Ambient temperature is used to account for seasonal thermal effects on PV output performance.

Three representative days have been selected to characterize seasonal variation in solar availability, corresponding to Summer, Winter, and Monsoon conditions. These seasonal profiles are used to simulate realistic PV generation behavior under varying climatic conditions throughout the year.

6.1.1 Seasonal Irradiance Profiles

The irradiance and temperature data for the three seasons reflect the distinct climatic conditions prevalent in Maharashtra:

Summer (March–May): High solar irradiance with clear skies and peak GHI values typically exceeding 800 W/m^2 . Ambient temperatures are also high, generally ranging between 35°C to 44°C . Although high temperatures slightly reduce PV module efficiency, the overall solar energy output remains maximum due to longer sunshine duration and strong irradiance availability.

Winter (November–January): Moderate irradiance with shorter daylight hours and lower sun angles. GHI values are lower compared to summer, typically ranging between $500\text{--}700\text{ W/m}^2$, while ambient temperatures remain relatively low between 15°C to 28°C . Lower temperatures improve PV module efficiency, partially compensating for the reduced irradiance.

Monsoon (June–September): Significantly reduced irradiance due to heavy cloud cover and frequent rainfall. GHI values often remain below $400\text{--}500\text{ W/m}^2$, and ambient temperatures range between 25°C to 32°C . Intermittent and diffuse irradiance dominates during this season, resulting in substantially lower PV generation. This represents the most challenging operating condition for solar-powered irrigation systems.

6.2 Agricultural Load Profile

The overall electricity demand curve for Maharashtra state 4 was used as the reference for developing the irrigation feeder load profile. Based on analysis of the Maharashtra state load data, agricultural consumption constitutes approximately 22% of the total electricity demand on an annual average basis. This percentage is consistent with reported figures for agriculturally intensive states in India, where irrigation pumping is a dominant end-use. To represent seasonal variation in agricultural demand, different seasonal scaling factors have been assigned to each season. These factors reflect the variation in irrigation requirements driven by crop cycles, rainfall patterns, and groundwater availability throughout the year. The seasonal fractions have been carefully chosen such that when averaged over the three seasons, the resulting agricultural load fraction equals the observed 22% annual average.

6.2.1 Seasonal Agricultural Load Fractions

The seasonal agricultural load fractions applied to the Maharashtra state demand curve are summarized in the table below:

Season	Agricultural Load Fraction	Period
Summer	0.30 (30%)	February – May
Winter	0.20 (20%)	October – January
Monsoon	0.15 (15%)	June – September

Table 6.1: Seasonal Agricultural Load Fractions for Maharashtra

6.2.2 Derivation of Average Agricultural Fraction

The seasonal fractions are weighted by the number of months in each season to verify that the annual average aligns with the observed 22% agricultural consumption.

Assuming equal weighting across the three seasons (4 months each, totaling 12 months of defined seasons):

$$\frac{0.30 + 0.20 + 0.15}{3} \approx 0.22$$

This confirms that the selected seasonal fractions are consistent with the observed annual agricultural electricity consumption share of 22% for Maharashtra state.

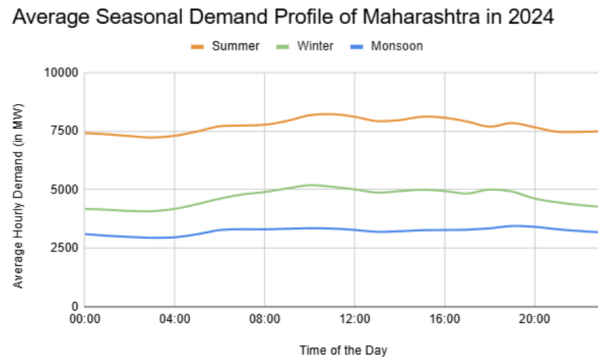


Figure 6.1: Seasonal Load Profile

Seasonal Agricultural Load Curves

The seasonal agricultural load curves are derived by multiplying the Maharashtra state **hourly demand curve** by the respective seasonal agricultural fraction. The resulting profiles represent the **net hourly power demand of the irrigation feeder** for each season (shown in Figure 6.1).

The Summer season exhibits the highest irrigation load, reflecting the peak agricultural water demand during the dry months when crops rely entirely on pump-based irrigation in the absence of monsoon rainfall. The Monsoon season shows the lowest agricultural electrical load, as rainfall partially or fully meets crop water requirements and farmers reduce pump operation. The Winter season represents an intermediate demand level.

These three seasonal load profiles are implemented in the simulation using a switching mechanism, allowing the system to alternate between seasonal demand conditions during time-series analysis. This enables a comprehensive assessment of the solar PV powered irrigation feeder performance across the full range of operating conditions experienced throughout the year.

6.2.3 Feeder Load Curve Development

Using the seasonal agricultural demand profiles, the feeder load curves are developed for simulation. The Maharashtra state hourly demand curve is first derived by multiplying the rated capacity of the pumping system (34.5kW) by the **variation fraction** (**The variation fraction is calculated by dividing hourly demand by the peak demand of the summer**) to obtain a per-unit load variation profile over a 24-hour period. This helps in preserving the shape of the actual demand pattern while allowing easy scaling according to the total load of the Feeder.

The normalized load curve is then multiplied by the corresponding seasonal agricultural load fraction for Summer, Winter, and Monsoon conditions. This produces three separate seasonal feeder load curves representing the actual irrigation pumping demand for each season.

The Summer load curve shows the highest demand due to intensive irrigation requirements during dry months, while the Monsoon load curve shows the lowest demand because rainfall reduces the dependency on electric pump operation. The Winter season represents moderate agricultural load demand.

These derived feeder load curves are used as input load profiles in the MATLAB/Simulink model through a switching mechanism, enabling dynamic seasonal power flow analysis of the solar PV powered irrigation feeder system.

Load Profiles

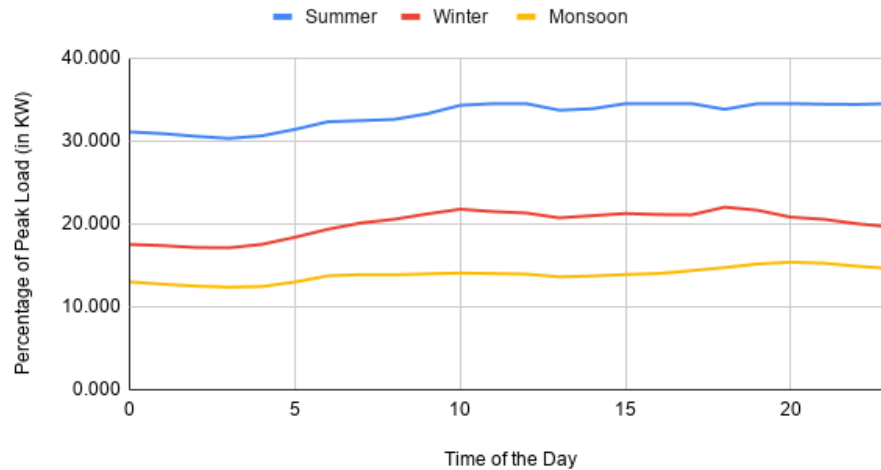


Figure 6.2: Seasonal Feeder Load Curves

Chapter 7

Power Flow and Simulation Results

The power flow simulation is carried out to evaluate the performance of the grid-connected solar PV powered irrigation feeder under different seasonal irradiance and agricultural load conditions. The analysis focuses on the interaction between solar PV generation, irrigation load demand, grid power support, and DC link voltage stability over a complete 24-hour operating cycle.

Separate simulations are performed for the three major agricultural seasons: Summer, Winter, and Monsoon. For each case, the corresponding seasonal irradiance profile and feeder load curve are applied in the MATLAB/Simulink model. The major output parameters observed are PV generated power, load consumption, grid power supply, and DC link voltage.

7.1 Summer Season Simulation

The Summer season represents the high irrigation demand and the maximum solar energy availability. The load consumption remains nearly constant throughout the day in the range of 32–34 kW due to intensive irrigation requirements during dry months.

The PV generation starts increasing after sunrise at around 6:30 AM, rises rapidly during the morning hours, and reaches a peak of nearly 36.5 kW around noon. During this period, the PV system supplies most of the feeder demand, significantly reducing the dependency on grid power. The grid power drops sharply during peak solar hours and reaches its minimum value around midday.

After sunset, PV generation decreases to zero and the grid again becomes the major source supplying the irrigation load. The DC link voltage remains regulated around the desired operating value with minor fluctuations and transient spikes, confirming stable operation of the boost converter and inverter under high irradiance conditions.

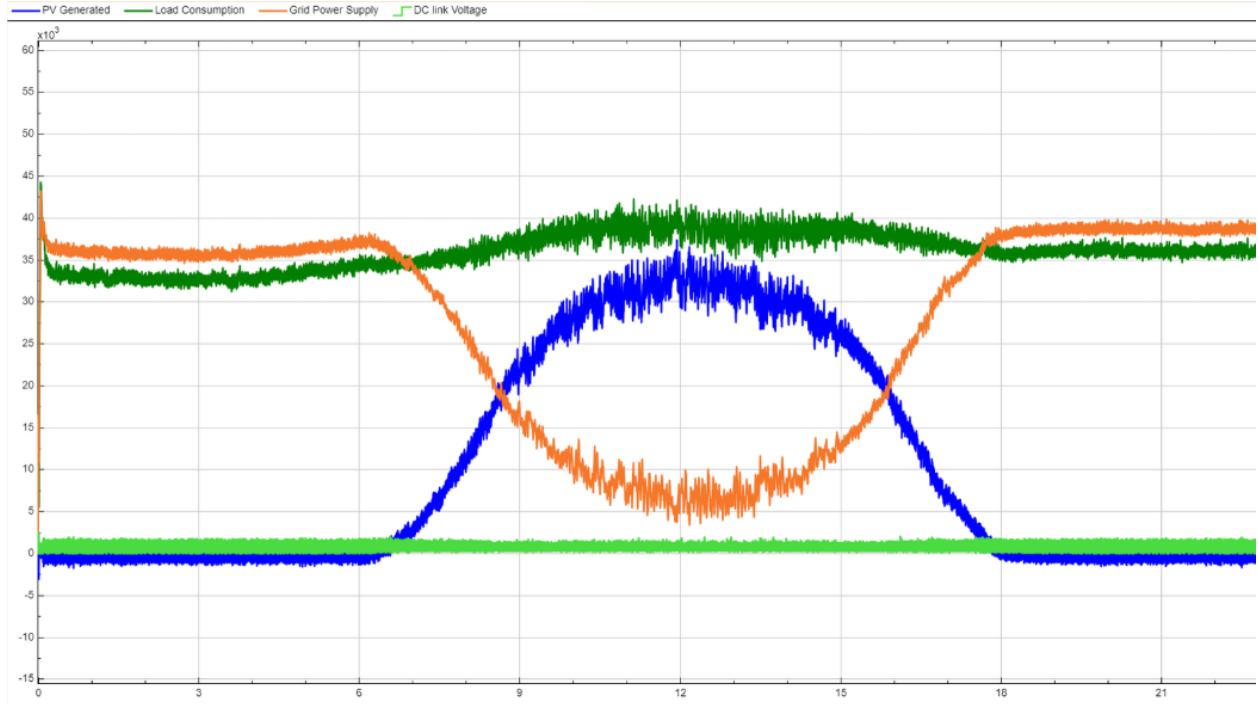


Figure 7.1: Power Flow Profile for Summer Season

7.2 Winter Season Simulation

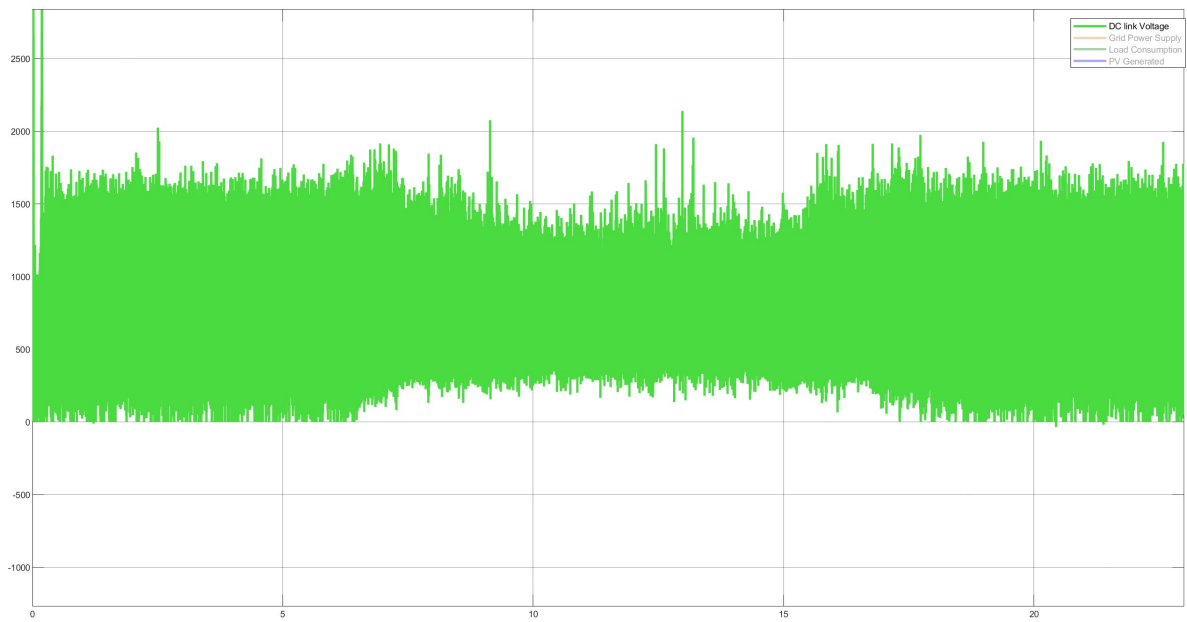


Figure 7.2: DC Link Voltage Profile for Winter Season

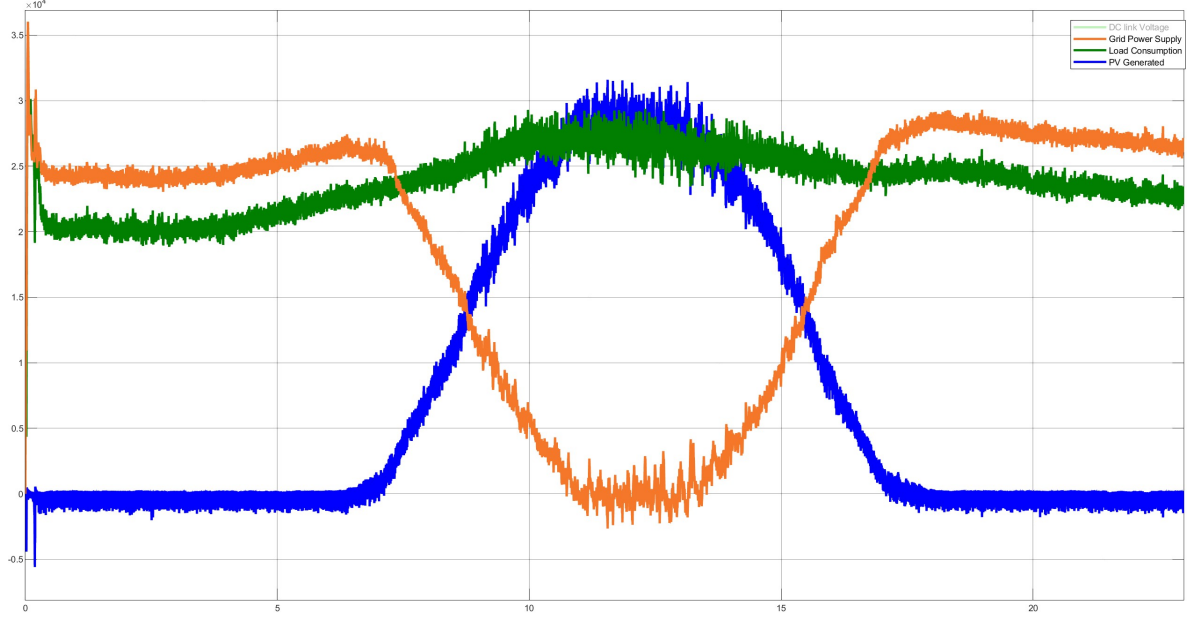


Figure 7.3: Power Flow Profile for Winter Season

The Winter season shows moderate solar irradiance and intermediate agricultural demand. The load consumption remains in the range of approximately 23–30 kW, which is lower than summer due to reduced irrigation requirements.

PV generation begins in the morning and gradually increases to a peak value of nearly 33 kW around noon. Since winter irradiance is lower than summer, the PV contribution is comparatively smaller. However, during midday hours, the PV system still supplies a major portion of the feeder load and substantially reduces the grid power requirement.

The grid supplies most of the demand during early morning and evening hours when PV generation is unavailable. Around noon, the grid power reaches its lowest value due to the maximum contribution from the PV system. The DC link voltage remains stable with small fluctuations, showing effective converter control and satisfactory system performance.

7.3 Monsoon Season Simulation

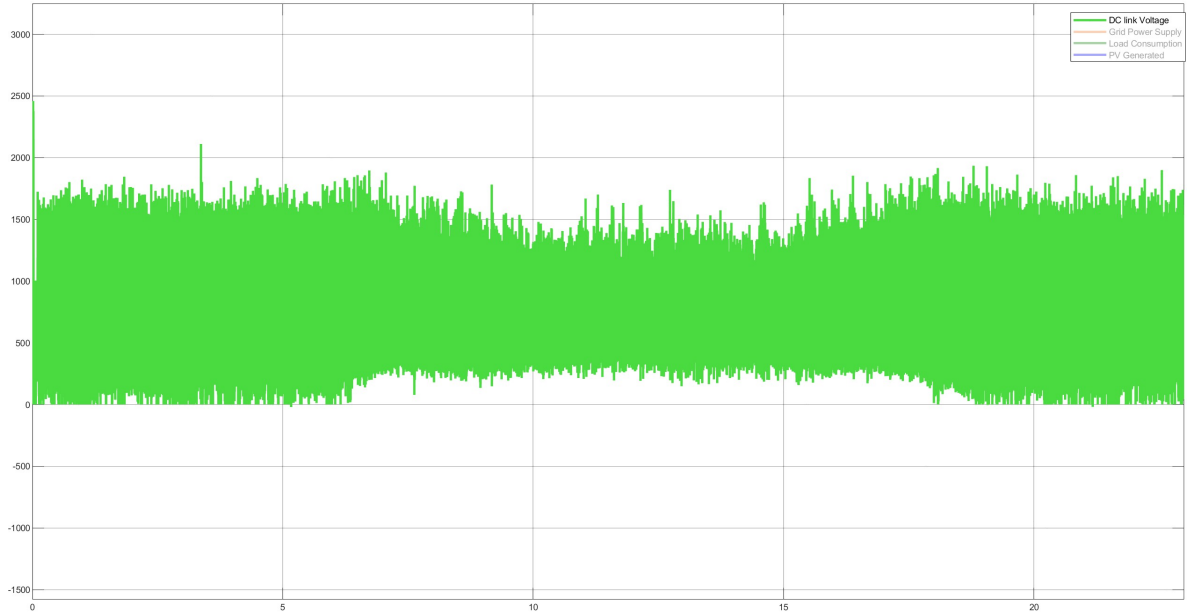


Figure 7.4: DC Link Voltage Profile for Monsoon Season

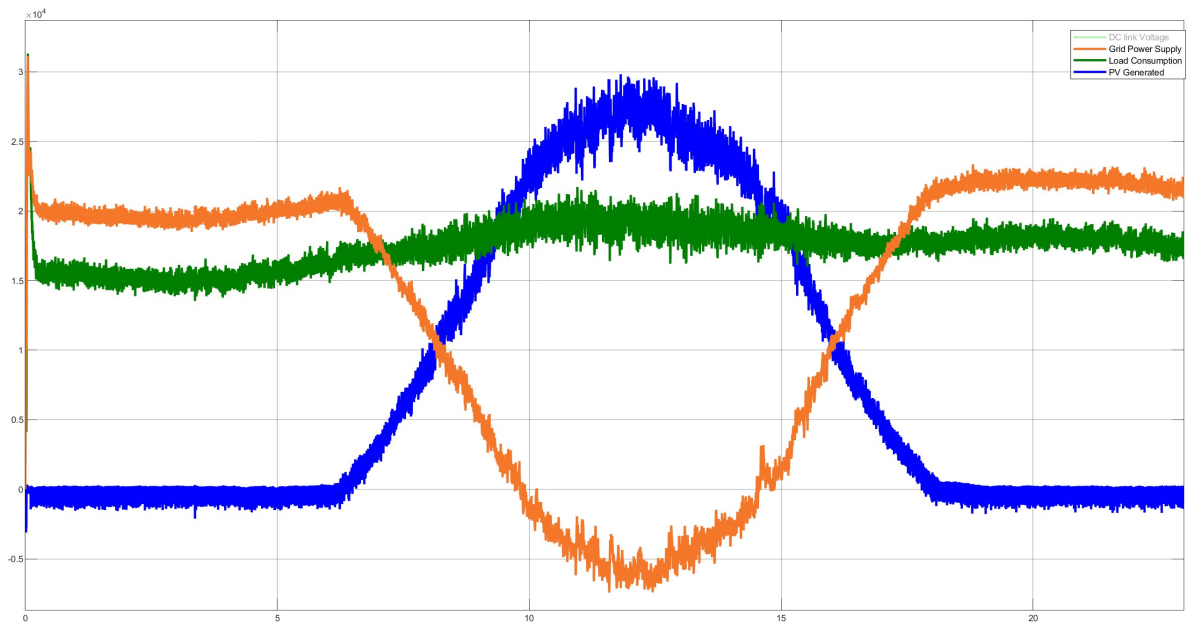


Figure 7.5: Power Flow Profile for Monsoon Season

In the Monsoon season load consumption remains nearly constant in the range of 15–20 kW, lower a bit because less water is required in the season for irrigation from pumps. PV generation starts after sunrise and reaches a maximum value of nearly 37.5 kW around noon.

As a result, grid support is primarily required during the morning and evening hours when PV generation is low. During daytime, as solar generation increases and exceeds the load demand, the grid power reduces

significantly and becomes negative, indicating export of excess power to the grid. The DC link voltage shows small variations due to changing irradiance conditions but remains within the acceptable operating range, ensuring stable system performance.

7.4 Overall Discussion

The simulation results clearly show the seasonal dependence of solar PV contribution in the irrigation feeder. Summer provides the highest PV generation due to strong solar irradiance and longer sunshine hours, while Monsoon provides the lowest PV contribution because of cloud cover and reduced sunlight availability. Winter represents an intermediate operating condition.

The grid acts as a balancing source in all three seasons. During peak solar hours, the PV system reduces the power drawn from the grid, whereas during morning, evening, and low irradiance conditions, the grid supplies the remaining feeder demand.

The DC link voltage remains well regulated in all seasonal cases, confirming the proper functioning of the MPPT controller, DC–DC boost converter, and voltage source inverter. The results demonstrate that solar PV integration with agricultural irrigation feeders can effectively reduce grid dependency, improve renewable energy utilization, and provide reliable support for irrigation pumping loads under realistic seasonal operating conditions.

Chapter 8

Observation & Conclusion

Key Observations:

- The significant error between the load consumption and grid power and the huge ripples are due to the use of lower step time (10^{-4}) to simulate the result. Using an appropriate (10^{-5}) will get more accurate result

The project successfully developed and analyzed a solar PV powered pumps, irrigation feeder system using MATLAB/Simulink. A reduced irrigation feeder model was selected and modified to make the simulation computationally feasible while preserving the important electrical characteristics of the feeder network.

A solar-powered irrigation pumping system was designed for a total DC capacity of 43.125 kW, which was further rounded to a 45 kW PV system. The system consists of 75 PV modules of 600 Wp each, arranged in a 15-series and 5-parallel configuration. Multiple agricultural pumps were integrated into the feeder, including 5 HP and 10 HP pumping loads, representing practical irrigation applications.

Seasonal 24-hour load curves were estimated for Summer, Winter, and Monsoon conditions using Maharashtra state demand data and seasonal agricultural load fractions. These curves were used in the simulation to study the performance of the feeder under different operating conditions.

The power flow simulation results show that the PV system supplies a significant portion of the irrigation load during solar hours, reducing the dependency on grid power. The grid acts as a balancing source when PV generation is insufficient, especially during morning, evening, and Monsoon conditions. The DC link voltage remains regulated in all seasonal cases, confirming stable operation of the boost converter and inverter control system.

Therefore, the major objectives of the project have been achieved:

- A real-life irrigation feeder was selected and reduced for simulation.
- 3 solar-powered irrigation pumping systems of 15 HP capacity was designed.
- Multiple pumps were integrated with the feeder and system performance was analyzed.
- Seasonal 24-hour load curves were estimated for Summer, Winter, and Monsoon.
- Load flow and simulation results were obtained for the developed system.
- A comprehensive project report was prepared covering feeder design, PV sizing, load profile development, and simulation results.

Overall, the study demonstrates that solar PV integration with agricultural irrigation feeders can reduce grid dependency and support sustainable irrigation practices. The developed model can be further extended for larger feeder networks, detailed voltage profile analysis, reactive power control, and economic feasibility assessment.

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